TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	81.3 ± 2.15
Correlation Stability	81.8 ± 0.60
Distribution Consistency	85.4 ± 1.80
Entropy Preservation	19.9 ± 0.83
Mean Z Score Drift	0.3 ± 0.06
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	84.1 ± 2.39
Novelty Score	0.9 ± 0.01

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	51.132	-3.307	6.1%	OK
sex	0.680	0.616	-0.064	9.4%	OK
cp	3.158	3.153	-0.006	0.2%	OK
trestbps	131.690	122.992	-8.698	6.6%	OK
chol	246.693	222.727	-23.966	9.7%	OK
fbs	0.149	0.207	+0.058	39.1%	Drift
restecg	0.990	0.880	-0.110	11.1%	Warn
thalach	149.607	129.112	-20.496	13.7%	Warn
exang	0.327	0.326	-0.000	0.1%	OK
oldpeak	1.040	1.615	+0.576	55.4%	Drift
slope	1.601	1.818	+0.218	13.6%	Warn
ca	0.672	0.813	+0.140	20.9%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	81.3 ± 2.15
Correlation Stability (*2)	81.8 ± 0.60
Distribution Consistency (*3)	85.4 ± 1.80
Entropy Preservation (*4)	19.9 ± 0.83
Mean Z Score Drift (*5)	0.3 ± 0.06
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	84.1 ± 2.39
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.5049180327868853	0.6049180327868853	-0.00327868852459016	
F1	0.6070236204175147	0.392104807986111	0.5595397132226607	-0.04748390719485389	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5901639344262295	0.5426229508196722	0.601639344262295	0.01147540983606560 3	
F1	0.5480779520963047	0.40602926984315635	0.5389079653267266	-0.00916998676957814 7	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5360655737704919	0.440983606557377	0.5737704918032788	0.03770491803278687 4	
F1	0.5231512298313296	0.3862816517980949	0.538582148778348	0.01543091894701840 6	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5180327868852459	0.4327868852459016	0.5639344262295082	0.04590163934426229	
F1	0.4782496199381977	0.37955171644802516	0.5058816632240888	0.02763204328589113	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5131147540983606	0.5836065573770491	-0.01475409836065574 9	
F1	0.5648347697855012	0.389258696562831	0.48650461738333417	-0.07833015240216692	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5475409836065573	0.5475409836065573	0	
F1	0.3899270600110827	0.3899270600110827	0.3899270600110827	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.4524590163934426	0.562295081967213	0.019672131147540968	
F1	0.5198013329305267	0.4002298638214071	0.5368958357880784	0.017094502857551584	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.3967213114754099	0.5901639344262295	0.032786885245901655	
F1	0.568660979990255	0.3464582003196773	0.5620463410007568	-0.006614638989498278	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5262295081967212	0.4032786885245915	0.5180327868852459	-0.008196721311475414	
F1	0.5272786716544238	0.3823337154681753	0.5110918957476365	-0.016186775906787243	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.5065573770491804	0.537704918032787	-0.01639344262295081	
F1	0.4442202071436568	0.3832304361293196	0.4256511046451316	-0.01856910249852513	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	58.0	0.0	4.0	123.0	200.0	0.0	0.0	76.0	0.0	1.1063829679245947
2	62.0	1.0	3.0	131.0	234.0	1.0	0.0	77.0	1.0	4.341904725046457
3	53.0	1.0	4.0	72.0	248.0	0.0	0.0	154.0	0.0	4.11380424030852
4	31.0	1.0	3.0	115.0	254.0	0.0	0.0	109.0	0.0	4.212349405729222
5	33.0	0.0	2.0	141.0	126.0	0.0	0.0	150.0	1.0	0.3426173777235316
6	43.0	1.0	4.0	87.0	236.0	0.0	0.0	165.0	0.0	4.299700340214938 5
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	46.0	1.0	3.0	101.0	203.0	0.0	0.0	191.0	0.0	0.6427423783974655
2	44.0	1.0	1.0	151.0	329.0	0.0	2.0	161.0	0.0	0.2439609890855305 4
3	41.0	1.0	4.0	103.0	170.0	0.0	2.0	158.0	1.0	0.6107843709572706
4	66.0	1.0	2.0	85.0	376.0	0.0	2.0	140.0	0.0	-0.05174670766634669
5	62.0	0.0	3.0	117.0	230.0	0.0	2.0	190.0	1.0	3.2138454571369244
6	47.0	0.0	4.0	85.0	260.0	0.0	0.0	126.0	1.0	0.3078705830038507
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	62.0	1.0	3.0	200.0	328.0	0.0	0.0	152.0	1.0	0.21680247528403582
2	39.0	1.0	4.0	103.0	173.0	0.0	2.0	145.0	1.0	0.3187730882965196
3	53.0	1.0	4.0	168.0	174.0	0.0	2.0	144.0	0.0	0.3144953818556736
4	48.0	1.0	3.0	153.0	444.0	0.0	0.0	166.0	0.0	0.4841138006225003
5	72.0	1.0	2.0	136.0	268.0	0.0	0.0	146.0	0.0	0.3849396260254708
6	75.0	0.0	2.0	114.0	207.0	1.0	0.0	164.0	0.0	4.607127451221929
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	1.0	4.0	170.0	364.0	1.0	0.0	123.0	1.0	-0.416762282688192
2	44.0	0.0	4.0	172.0	299.0	1.0	0.0	184.0	0.0	-0.29506585492625387
3	60.0	1.0	4.0	167.0	387.0	0.0	2.0	158.0	1.0	-0.026498912638412231
4	41.0	1.0	3.0	164.0	234.0	1.0	2.0	123.0	1.0	0.16541435013389472
5	74.0	0.0	4.0	169.0	256.0	1.0	2.0	161.0	0.0	1.586447514851335
6	56.0	0.0	3.0	179.0	382.0	0.0	0.0	164.0	1.0	4.65556355664084
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	47.0	1.0	4.0	135.0	188.0	0.0	0.0	104.0	0.0	1.6612401984776066
2	51.0	0.0	1.0	155.0	159.0	0.0	2.0	195.0	0.0	2.2661732690689543
3	74.0	1.0	1.0	144.0	152.0	0.0	0.0	199.0	1.0	1.9367025734681007
4	38.0	0.0	3.0	156.0	193.0	0.0	0.0	160.0	0.0	0.7037300512089912
5	59.0	0.0	4.0	117.0	131.0	0.0	2.0	174.0	0.0	0.041804365868842105
6	70.0	1.0	1.0	146.0	263.0	0.0	0.0	185.0	0.0	1.4171252071287856
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	71.0	1.0	4.0	157.0	303.0	1.0	0.0	133.0	1.0	2.2484188547781123
2	66.0	1.0	4.0	189.0	224.0	0.0	2.0	93.0	1.0	2.7540412017373304
3	77.0	1.0	1.0	88.0	230.0	0.0	2.0	161.0	0.0	0.38258868909912164
4	54.0	1.0	3.0	140.0	314.0	0.0	0.0	148.0	1.0	1.7886649073781489
5	72.0	0.0	4.0	147.0	190.0	0.0	2.0	133.0	1.0	2.6589218221197806
6	60.0	0.0	1.0	148.0	280.0	0.0	0.0	133.0	1.0	1.9880108448428495
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	68.0	0.0	4.0	99.0	122.0	0.0	0.0	174.0	0.0	0.61207476427975 2
2	65.0	1.0	4.0	139.0	246.0	0.0	0.0	143.0	0.0	5.99968250084061 2
3	52.0	0.0	3.0	120.0	276.0	0.0	2.0	133.0	0.0	6.09790196414940 2
4	57.0	1.0	4.0	116.0	415.0	0.0	2.0	181.0	0.0	2.02680760127608 1
5	79.0	1.0	3.0	152.0	244.0	0.0	2.0	125.0	1.0	0.87780549083629 07
6	63.0	0.0	4.0	159.0	204.0	0.0	2.0	154.0	0.0	1.42045541949247 63
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	29.0	0.0	2.0	161.0	261.0	0.0	0.0	133.0	1.0	0.40461337429187 27
2	40.0	1.0	1.0	149.0	249.0	0.0	2.0	116.0	0.0	0.43577950021791 58
3	58.0	1.0	3.0	173.0	207.0	0.0	2.0	90.0	0.0	0.06150819466122 378
4	56.0	0.0	4.0	145.0	176.0	0.0	2.0	101.0	1.0	0.48566540151190 04
5	58.0	1.0	2.0	160.0	230.0	0.0	0.0	130.0	1.0	0.01484984468468 5247
6	64.0	1.0	4.0	185.0	161.0	0.0	1.0	145.0	0.0	0.45651772975468 585
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	46.0	1.0	4.0	131.0	278.0	0.0	2.0	203.0	0.0	0.68769667160692 3
2	38.0	1.0	4.0	128.0	306.0	0.0	0.0	178.0	0.0	1.39263178165133 05
3	46.0	1.0	4.0	116.0	284.0	0.0	0.0	167.0	1.0	5.24760314475893 5
4	40.0	1.0	4.0	110.0	307.0	1.0	2.0	143.0	1.0	4.37105581469641 2
5	48.0	0.0	4.0	105.0	214.0	0.0	2.0	144.0	1.0	1.88859784961901 54
6	45.0	1.0	4.0	121.0	188.0	0.0	2.0	160.0	0.0	0.72912106056274 66
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	32.0	1.0	4.0	127.0	262.0	0.0	2.0	156.0	0.0	1.73293826461573 9
2	55.0	1.0	4.0	161.0	128.0	0.0	2.0	164.0	1.0	-0.48399556974037 95
3	66.0	1.0	4.0	139.0	170.0	0.0	2.0	180.0	0.0	-0.26906297611510 49
4	50.0	1.0	3.0	192.0	138.0	0.0	0.0	105.0	0.0	0.00586208695332 0578
5	60.0	0.0	1.0	138.0	272.0	0.0	2.0	128.0	0.0	-0.13331424901756 583
6	68.0	0.0	2.0	163.0	156.0	1.0	2.0	182.0	0.0	0.15931796781761 22
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6